Module 3 and 4 assessment questions.

Tools you can use for below questions: VirtualBox with Linux VMs, GNS3, Cisco Packet Tracer, Wireshark etc.

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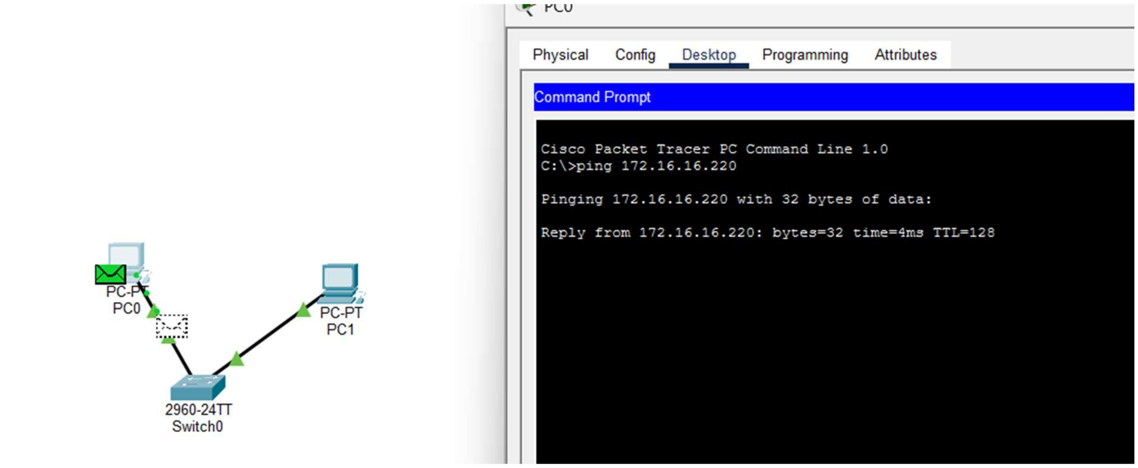
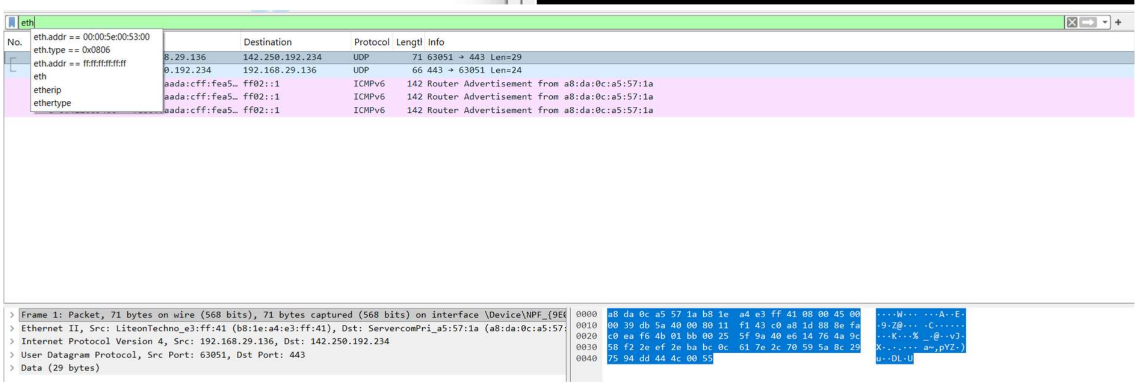
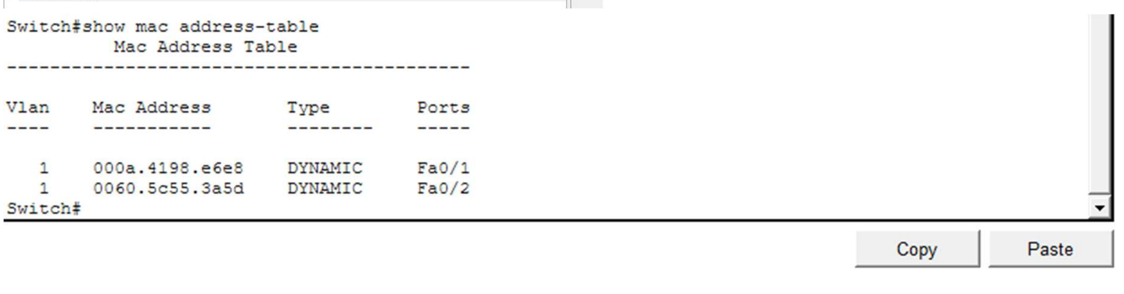
- Simulate a small network with switches and multiple devices. Use ping to generate traffic and observe the MAC address table of the switch. Capture packets using Wireshark to analyze Ethernet frames and MAC addressing.
- Capture and analyze Ethernet frames using Wireshark. Inspect the structure of the frame, including destination and source MAC addresses, Ethertype, payload, and FCS. Use GNS3 or Packet Tracer to simulate network traffic.
- Configure static IP addresses, modify MAC addresses, and verify network connectivity using ping and ifconfig commands.
- Troubleshoot Ethernet Communication with ping and traceroute -> Using cisco packet tracer:
- Create a simple LAN setup with two Linux machines connected via a switch.
- Ping from one machine to the other. If it fails, use ifconfig to ensure the IP addresses are configured correctly.
- Use traceroute to identify where the packets are being dropped if the ping fails.
- Research the Linux kernel's handling of Ethernet devices and network interfaces. Write a short report on how the Linux kernel supports Ethernet communication (referencing kernel.org documentation).
- Describe how you would configure a basic LAN interface using the ip command in Linux (kernel.org).
- Use Linux to view the MAC address table of a switch (if using a Linux-based network switch). Use the bridge or ip link commands to inspect the MAC table and demonstrate a basic switch's operation.
- Submit your findings with explanations of what you observe.

1.

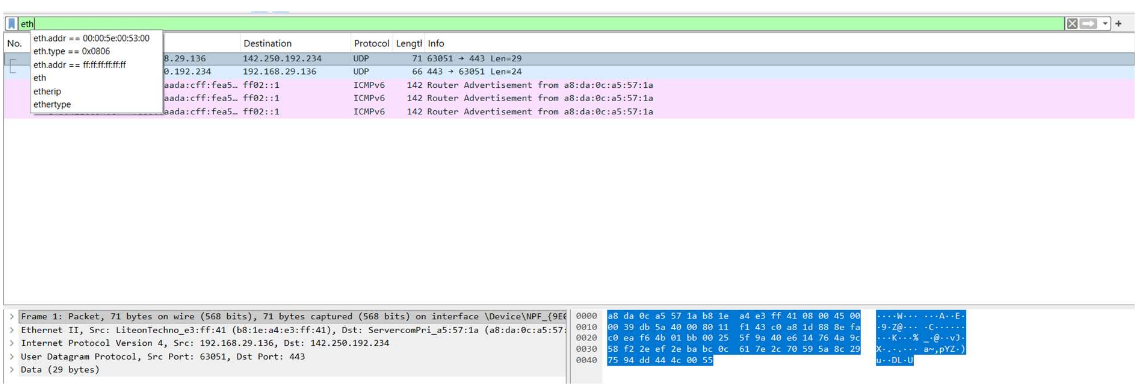

PC0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.16.16.220
Pinging 172.16.16.220 with 32 bytes of data:
Reply from 172.16.16.220: bytes=32 time=4ms TTL=128
PC-PT
PC0
PC-PT
PC1
2960-24TT
Switch0


eth
eth.addr == 00:00:5e:00:53:00
eth.type == 0x0806
eth.addr == ff:ff:ff:ff:ff:ff
eth
etherip
ethertype
No.
Destination
Protocol
Length
Info
142.250.192.234 UDP 71 63051 → 443 Len=29
192.168.29.136 UDP 66 443 → 63051 Len=24
ff02::1 ICMPv6 142 Router Advertisement from a8:da:0c:a5:57:1a
ff02::1 ICMPv6 142 Router Advertisement from a8:da:0c:a5:57:1a
ff02::1 ICMPv6 142 Router Advertisement from a8:da:0c:a5:57:1a
Frame 1: Packet, 71 bytes on wire (568 bits), 71 bytes captured (568 bits) on interface \Device\NPF_{9E
Ethernet II, Src: LiteonTechno_e3:ff:41 (b8:1e:a4:e3:ff:41), Dst: ServercomPri_a5:57:1a (a8:da:0c:a5:57:
Internet Protocol Version 4, Src: 192.168.29.136, Dst: 142.250.192.234
User Datagram Protocol, Src Port: 63051, Dst Port: 443
Data (29 bytes)


Switch#show mac address-table
Mac Address Table

Vlan Mac Address Type Ports
---- ----------- -------- -----
1 000a.4198.e6e8 DYNAMIC Fa0/1
1 0060.5c55.3a5d DYNAMIC Fa0/2
Switch#
Copy
Paste

2.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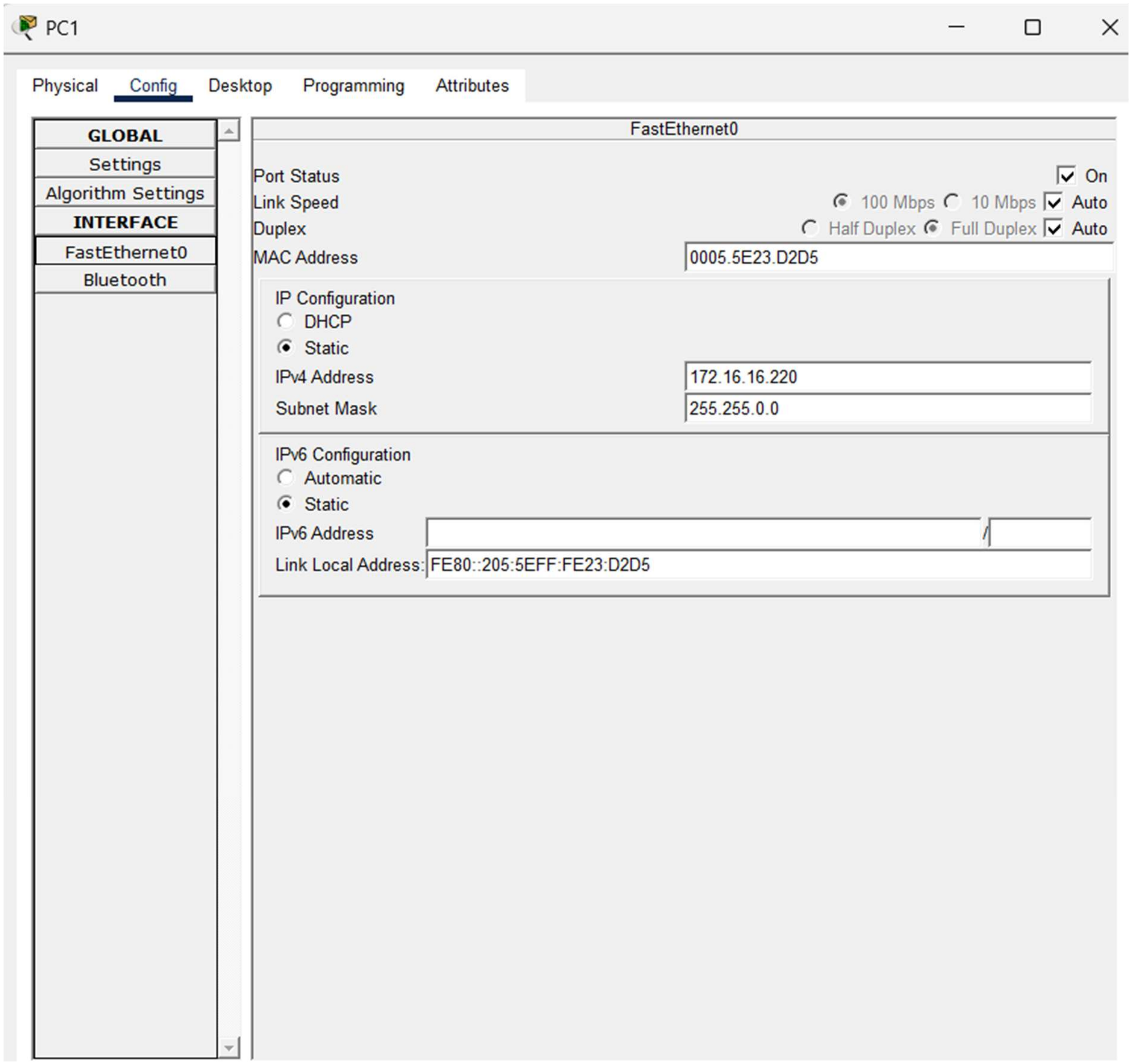
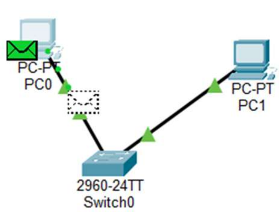
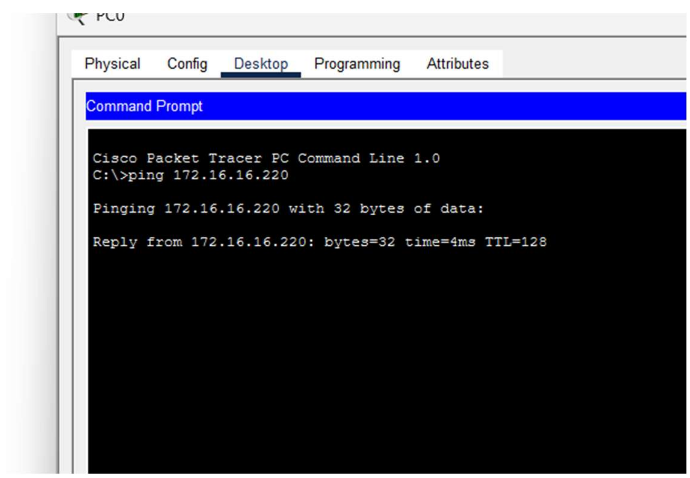
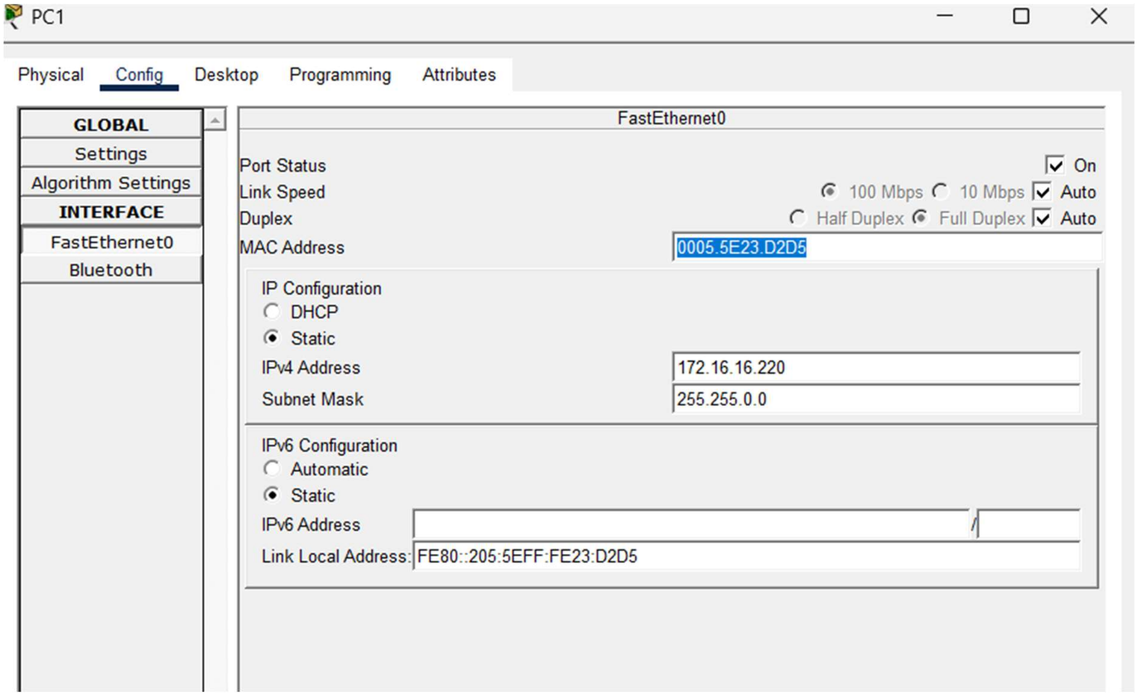
User Datagram Protocol, Src Port: 63051, Dst Port: 443
Data (29 bytes)

3.


PC1
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
FastEthernet0
Port Status
On
Link Speed
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0005.5E23.D2D5
IP Configuration
DHCP
Static
IPv4 Address
172.16.16.220
Subnet Mask
255.255.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address:
FE80::205:5EFF:FE23:D2D5


PC0
PC-PT
PC1
2960-24TT
Switch0


Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.16.16.220
Pinging 172.16.16.220 with 32 bytes of data:
Reply from 172.16.16.220: bytes=32 time=4ms TTL=128



4.

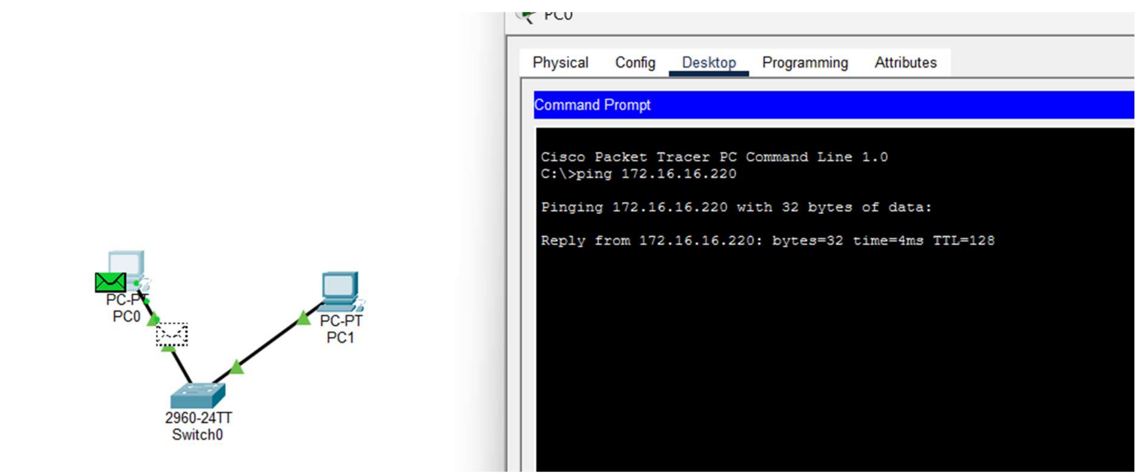
```
C:\>tracert 172.16.16.220

Tracing route to 172.16.16.220 over a maximum of 30 hops:

  1   4 ms      4 ms      4 ms      172.16.16.220

Trace complete.
```

5.



6.

```
C:\>ping 172.16.16.200

Pinging 172.16.16.200 with 32 bytes of data:

Reply from 172.16.16.200: bytes=32 time=1ms TTL=128
Reply from 172.16.16.200: bytes=32 time=2ms TTL=128
Reply from 172.16.16.200: bytes=32 time=4ms TTL=128
Reply from 172.16.16.200: bytes=32 time=1ms TTL=128

Ping statistics for 172.16.16.200:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 4ms, Average = 2ms

C:\>ifconfig
Invalid Command.

C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::20A:41FF:FE85:5357
   IPv6 Address....................: ::
   IPv4 Address....................: 172.16.16.200
   Subnet Mask.....................: 255.255.0.0
   Default Gateway.................: ::
                                     0.0.0.0

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>
```

Linux Kernel's Handling of Ethernet Devices and Network Interfaces

The Linux kernel manages Ethernet devices via the **netdev** subsystem, enabling communication through network drivers and protocols. It supports interface management using ip link, packet transmission via the **socket API**, and device configurations via /sys/class/net/. Advanced features like **eBPF** allow efficient packet filtering and monitoring

To configure a LAN interface in Linux, first **bring up the interface** using ip link set eth0 up. Assign an IP address with ip addr add 192.168.1.100/24 dev eth0, then verify with ip addr show eth0. For persistence, configurations should be added to /etc/network/interfaces or netplan.